

Kshitiz Neupane

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EDUCATION

University of Southern Mississippi

B.S. in Computer Science and Mathematics; GPA: 4.0 / 4.0

Hattiesburg, MS

Expected May 2028

Coursework: Data Structures, Linear Algebra, Probability & Statistics, Differential Equations

EXPERIENCE

Research Assistant — ML, NLP & Cybersecurity

University of Southern Mississippi

Aug. 2024 – Present

Hattiesburg, MS

- Developed a two-stage email-analysis pipeline combining graph-neural-network anomaly detection with ModernBERT content verification; reported 86% filtering recall and over 92% precision after verification.
- Automated data preparation, tokenization, GPU-aware training, and evaluation with PyTorch/PyG, Transformers, and scikit-learn; documented baseline comparisons, diagnostic visualizations, and failure modes.

Software Developer Intern

Butwal Technologies

Feb. 2024 – Aug. 2024

Butwal, Nepal

- Created test scripts and technical reports for API and automation work; collaborated in agile development and maintained documentation to make testing repeatable.

SELECTED PROJECTS

ShiftGuard — Model Failure Under Distribution Shift

GitHub

Python, SciPy, scikit-learn

- Built a reproducible benchmark spanning 3 datasets, 5 classifiers, and 5 seeds, producing 12,000 model-window evaluations to compare Wasserstein, KL, PSI, and MMD as label-free failure signals.
- Measured an overall failure ROC-AUC of 0.799 for maximum marginal Wasserstein (95% CI: 0.760–0.832); used validation-only calibration and documented variation across datasets and shift types.

WorkForge AI — Retrieval-Augmented Project Management

React, FastAPI, ChromaDB

- Built a React/FastAPI platform with local retrieval-augmented generation using ChromaDB and sentence-transformers; implemented PDF, DOCX, Markdown, and TXT ingestion with chunking and persistent vector storage.
- Developed a five-step agent workflow to retrieve deliverables and project context, analyze requirements, suggest improvements, and compile review reports; added approvals and PDF proposal generation.

ManageYourHome — Household Expense Sharing

GitHub

React, TypeScript, Django REST, JWT

- Built a full-stack expense-sharing app with JWT access-token refresh, authenticated CRUD APIs, pagination, and server-side per-user expense and tax calculations.

F1 Race Analysis & Predictive Modeling

GitHub

Python, pandas, scikit-learn

- Engineered pre-race features and reusable preprocessing/model pipelines, achieving holdout podium ROC-AUC of 0.943 and finish-position MAE of 3.554; excluded in-race variables from predictive inputs.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++

ML & Data: PyTorch/PyG, Transformers, scikit-learn, pandas, NumPy, SciPy, NLP, Bayesian inference

Software & Tools: React, Django/DRF, FastAPI, REST APIs, Git, Linux, Docker, GitHub Actions, SQLite, PostgreSQL

Research Interest: Mechanistic interpretability; neural representations, circuits, and model reliability.